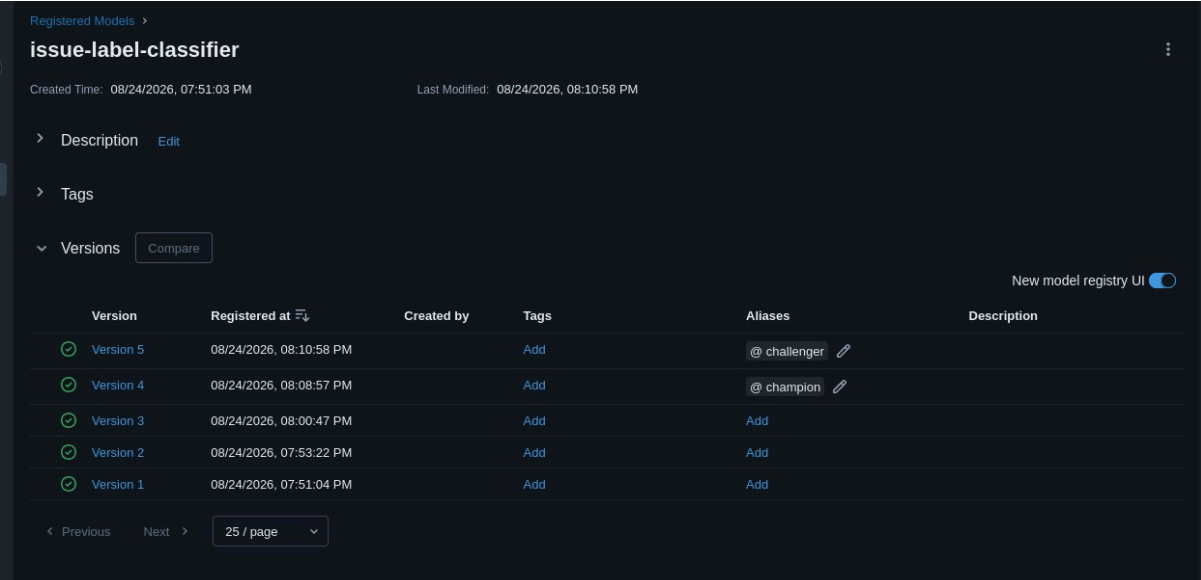


Assignment 3

This assignment extends the Assignment 2 pipeline with experiment tracking and a champion-challenger model promotion scheme, using a self-hosted MLflow server. Every training run, regardless of which model architecture produced it, is logged to MLflow with its parameters, metrics, and model artifact, then automatically evaluated against the current production model to decide whether it should replace it.

Champion-challenger logic: every run is registered in MLflow and tagged challenger. It's promoted to champion only if its micro-F1 beats the current champion's by a defined margin (0.005) or immediately if no champion exists yet. This runs identically regardless of which script produced the run, so neither model is favored by the mechanism itself; whichever is actually better wins.

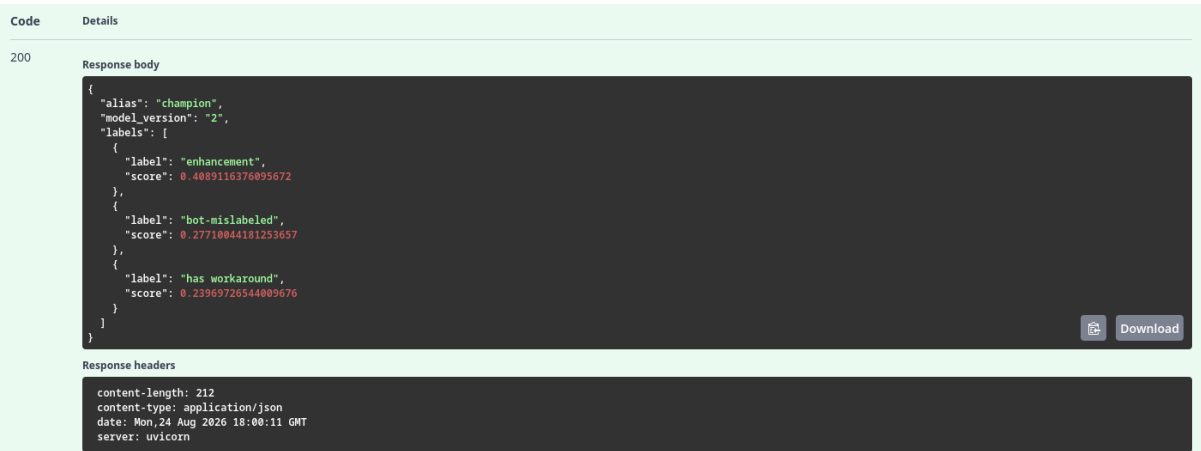
Here's challenger-champion models:



The screenshot shows the MLflow 'Registered Models' page for a model named 'issue-label-classifier'. It displays a table of model versions, with Version 5 being the current champion and Version 4 being the challenger. The table includes columns for Version, Registered at, Created by, Tags, Aliases, and Description. A 'Compare' button is visible next to the Versions section.

Version	Registered at	Created by	Tags	Aliases	Description
Version 5	08/24/2026, 08:10:58 PM		Add	@ challenger	
Version 4	08/24/2026, 08:08:57 PM		Add	@ champion	
Version 3	08/24/2026, 08:00:47 PM		Add	Add	
Version 2	08/24/2026, 07:53:22 PM		Add	Add	
Version 1	08/24/2026, 07:51:04 PM		Add	Add	

Champion:



The screenshot shows the MLflow Model Registry page for the 'Champion' model. It displays the response body and response headers for a GET request to the model's endpoint. The response body is a JSON object containing the model's alias, version, and labels with their scores.

Code	Details
200	Response body <pre>{ "alias": "champion", "model_version": "2", "labels": [{ "label": "enhancement", "score": 0.4089116376095672 }, { "label": "bot-misabeled", "score": 0.27710044181253657 }, { "label": "has workaround", "score": 0.23969726544009676 }]}</pre> Response headers <pre>content-length: 212 content-type: application/json date: Mon, 24 Aug 2026 18:00:11 GMT server: uvicorn</pre>

Challenger:

CodeDetails

200

Response body

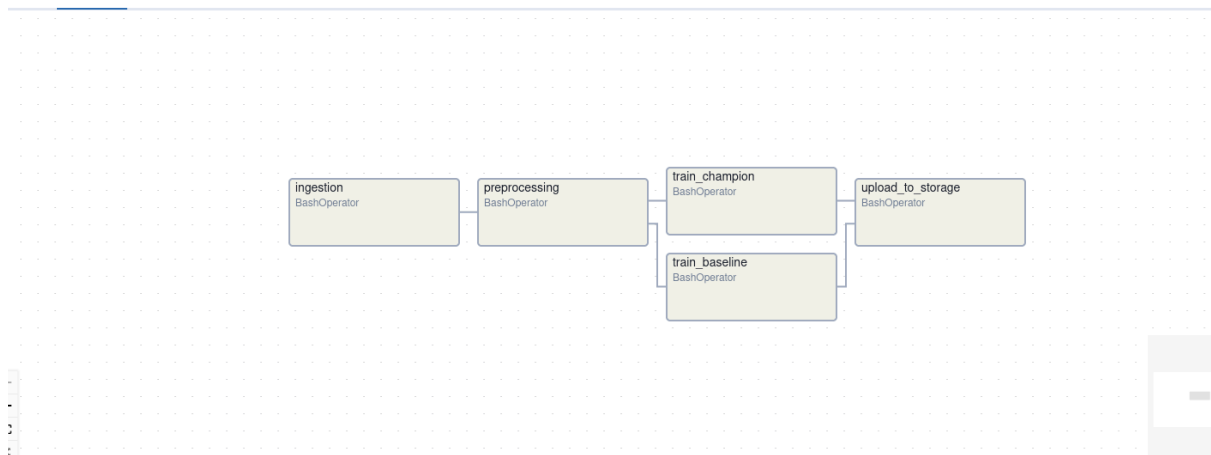
```
{
  "alias": "challenger",
  "model_version": "3",
  "labels": [
    {
      "label": "enhancement",
      "score": 0.34011630535750176
    },
    {
      "label": "bot-mislabeled",
      "score": 0.31572775865757613
    },
    {
      "label": "has workaround",
      "score": 0.31427481031021465
    }
  ]
}
```

Response headers

```
content-length: 215
content-type: application/json
date: Mon, 24 Aug 2026 18:01:45 GMT
server: uvicorn
```

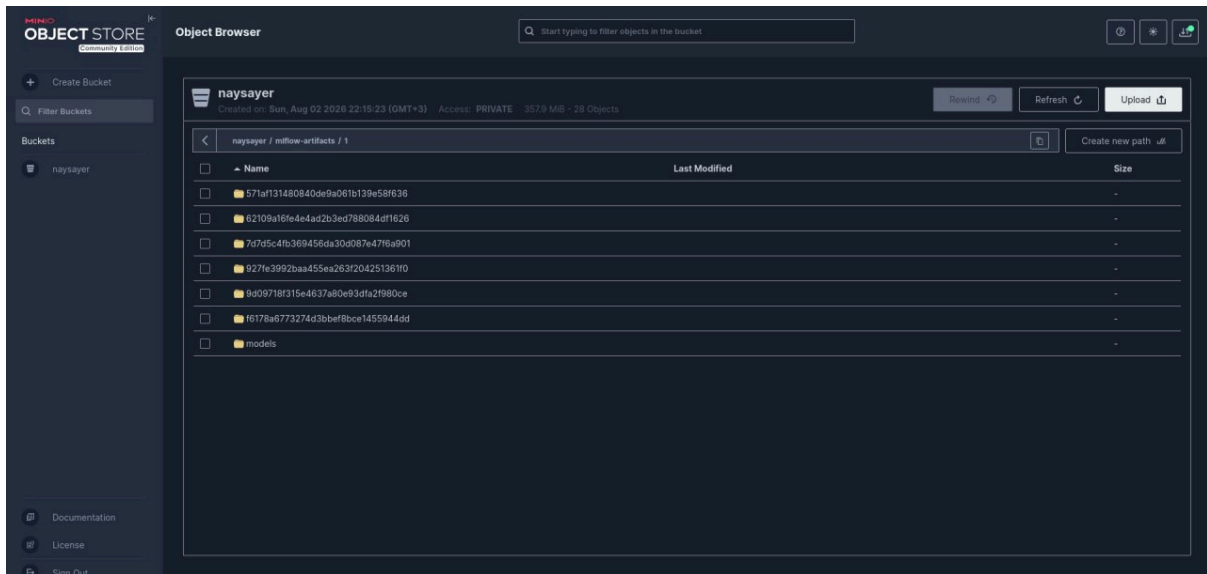
Download

As we can see, even though the predicted labels are similar, their predicted scores are different.

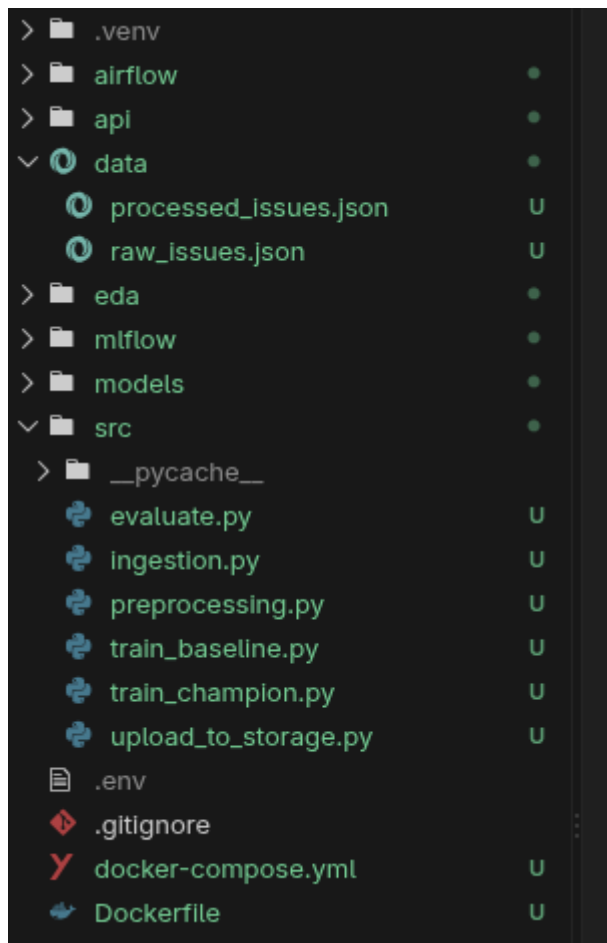


Here's our changed train files to imitate improvement

Here's our stored artifacts:



Here's projects overall structure:



Github: <https://github.com/GrimLord16/naysayer>